| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **7.** | (a) | (i) | | 30 [m/s] | | 1 | | 1 | 1 | |
| | | (ii) | | Substitution: $\frac{30\ \textbf{ecf}}{1.5}$ (1)<br>= 20 [m/s$^2$] (1) | 1 | 1 | | 2 | 2 | |
| | (b) | (i) | | **Indicative content**<br>AB – During the first 2.5 s there is uniform / constant acceleration from 0 to 40 m/s.<br>BC – Between 2.5 and 3 s there is uniform / constant deceleration from 40 to 20 m/s.<br>CD – After 3 s the car travels at constant speed of 20 m/s for 1 second.<br><br>**5–6 marks**<br>Comprehensively describes all three parts of the motion in detail and includes all values relevant to the motion.<br>*There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.*<br><br>**3–4 marks**<br>Comprehensively describes two of the parts of the motion in detail with some values **or** limited description of all three parts with some values included.<br>*There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* | 3 | 3 | | 6 | 3 | |

| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | | | | **1–2 marks**<br>Gives a limited description of one or two parts of the motion **or** description given of all parts with no data included.<br>*There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure*.<br>*The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.*<br><br>**0 marks**<br>*No attempt made or no response worthy of credit.* | | | | | | |
| | | (ii) | | $\text{speed} = \frac{\text{distance}}{\text{time}}$ or implied (1)<br>Substitution: $\text{speed} = \frac{85}{4}$ (1)<br>Speed = 21.25 or 21.3 or 21 [m/s] (1) | 1 | 1<br>1 | | 3 | 3 | |
| | | | | **Question 7 total** | **5** | **7** | **0** | **12** | **9** | **0** |